

Hadoop Multinode Cluster (Ubuntu)

1. Prepare the Nodes

- **Nodes:** At least 1 master (NameNode) and 2+ workers (DataNodes).
- **OS:** Ubuntu 20.04 or later.
- **Networking:** All nodes must be able to resolve each other's hostnames.
- Install Ubuntu (same version) on all machines.
- Assign **static IP addresses** and unique hostnames (e.g., node1, node2, node3).
- Ensure passwordless SSH between nodes:
- Generate the key while logging as the namenode and copy the generated key into data nodes and save. So that, it will give the access to the namenode to directly access into the data nodes without the use of entering the individual passwords.
- On the **master node**:

```
ssh-keygen -t rsa  
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_keys
```

- Copy the public key to each worker:

```
ssh-copy-id hadoop@dn1  
ssh-copy-id hadoop@dn2
```

2. Install the JAVA and HADOOP pre-requisites.

(a) Install Java

Hadoop requires Java to work.

```
sudo apt update  
sudo apt install openjdk-11-jdk -y  
java -version
```

(b) Download and Extract Hadoop

On all the nodes:

```
wget https://downloads.apache.org/hadoop/common/hadoop-3.3.6/hadoop-3.3.6.tar.gz
```

```
sudo tar -xzvf hadoop-3.3.6.tar.gz -C /usr/local/
```

```
sudo mv /usr/local/hadoop-3.3.6 /usr/local/hadoop
```

(c) Set the environment variables in the bash.

```
#HADOOP AND JAVA PATH
```

```
export HADOOP_HOME=/usr/local/hadoop
```

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
```

```
export PATH=$PATH:$HADOOP_HOME/bin:$HADOOP_HOME/sbin
```

3. Configure Hadoop Core Files.

Edit configs in \$HADOOP_HOME/etc/hadoop/:

- **core-site.xml:**

```
<configuration>
  <property>
    <name>fs.defaultFS</name>
    <value>hdfs://nn1:8020</value>
  </property>
  <property>
    <name>hadoop.tmp.dir</name>
    <value>/data/hdfs/tmp</value>
  </property>
  <property>
    <name>io.file.buffer.size</name>
    <value>131072</value>
  </property>
</configuration>
```

- **hdfs-site.xml:**

- This file handles how data is saved. On a worker node (dn2), the dfs.datanode.data.dir is the most important setting.

```
<configuration>
  <property>
    <name>dfs.namenode.name.dir</name>
```

```

        <value>/home/hadoop/hadoop/hdfs/nn</value>
    </property>
    <property>
        <name>dfs.replication</name>
        <value>2</value>
    </property>
    <property>
        <name>dfs.permissions.enabled</name>
        <value>>false</value>
    </property>
</configuration>

```

- **mapred-site.xml:**

- This file tells Hadoop that MapReduce should use the YARN framework.
- **Location:** /home/hadoop/hadoop-3.3.6/etc/hadoop/mapred-site.xml
- Open the file and add these properties:

```

<configuration>
    <property>
        <name>mapreduce.framework.name</name>
        <value>yarn</value>
    </property>
    <property>
        <name>yarn.app.mapreduce.am.env</name>
        <value>HADOOP_MAPRED_HOME=/home/hadoop/hadoop</value>
    </property>
    <property>
        <name>mapreduce.map.env</name>
        <value>HADOOP_MAPRED_HOME=/home/hadoop/hadoop</value>
    </property>
    <property>
        <name>mapreduce.reduce.env</name>
        <value>HADOOP_MAPRED_HOME=/home/hadoop/hadoop</value>
    </property>
    <property>
        <name>mapreduce.application.classpath</name>
        <value>$HADOOP_HOME/share/hadoop/mapreduce/*:$HADOOP_HOME/s
hare/hadoop/mapreduce/lib/*</value>
    </property>

```

</configuration>

- **yarn-site.xml:**

<!-- Site specific YARN configuration properties -->

<property>

<name>yarn.resourcemanager.hostname</name>

<value>nn1</value>

</property>

<property>

<name>yarn.nodemanager.aux-services</name>

<value>mapreduce_shuffle</value>

</property>

<property>

<name>yarn.log-aggregation-enable</name>

<value>true</value>

</property>

</configuration>

4. Configure Workers

Edit \$HADOOP_HOME/etc/hadoop/workers on the master:

dn1

dn2

5. Format HDFS

On the master:

hdfs namenode -format

6. Start Hadoop Services

On the master:

start-dfs.sh

start-yarn.sh

Check running daemons:

Jps

You should see **NameNode, DataNode, ResourceManager, NodeManager**.

OUTPUT:

C:\Windows\System32>jps

1268 DataNode

17108 NodeManager

6680 NameNode

21660 ResourceManager

25500 SecondaryNameNode

18492 Jps

7. Verify Installation

Check the UI ports whether the nodes are running successfully or not.